Angiosperms). Pollination in gymnosperms is said to be direct as the pollens are deposited directly on the exposed ovules, whereas in angiosperms it is said to be indirect, as the pollens are deposited on the stigma of the pistil. In majority of angiosperms, the flower opens and exposes its mature anthers and stigma for pollination. Such flowers are called **chasmogamous** and the phenomenon is **chasmogamy.** In other plants, pollination occurs without opening and exposing their sex organs. Such flowers are called **cleistogamous** and the phenomenon is **cleistogamy.**

Based upon the flower on which the pollen of a flower reaches, the pollination is classified into two kinds, namely, **self-pollination (Autogamy) and cross-pollination(Allogamy).**

A. Self-pollination or Autogamy (Greek Auto = self, gamos = marriage):

According to a majority of Botanists, the transfer of pollen on the stigma of the same flower is called **self-pollination or Autogamy**. Self-pollination is possible only in those plants which bear bisexual flowers. In order to promote self- pollination the flowers of the plants have several adaptations or mechanisms. They are:

1. Cleistogamy: In cleistogamy (Greek Kleisto = closed. Gamos = marriage) flowers never open and expose the reproductive organs and thus the pollination is carried out within the closed flower. *Commelina, Viola*, *Oxalis* are some examples for cleistogamous flowers. In *Commelina benghalensis,* two types of flowers are produced- aerial and underground flowers. The aerial flowers are brightly coloured, chasmogamous and insect pollinated. The underground flowers are borne on the subterranean branches of the rhizome that are dull, cleistogamous and self pollinated and are not dependent on pollinators for pollination.(Figure 1.11).

2. Homogamy: When the stamens and stigma of a flower mature at the same time it is said to be homogamy. It favours self-pollination to occur. Example: *Mirabilis jalapa, Catharanthus roseus*

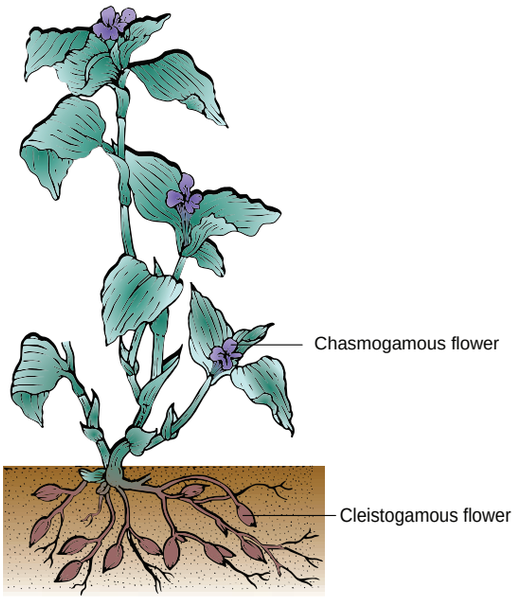


Figure 1.11 *Commelina* with Cleistogamous and Chasmogamous flowers

3. Incomplete dichogamy: In dichogamous flowers the stamen and stigma of a flower mature at different time. Sometimes , the time of maturation of these essential organs overlap so that it becomes favourable for self-pollination.

B. Cross - pollination

It refers to the transfer of pollens on the stigma of another flower. The cross-pollination is of two types:

i. Geitonogamy: When the pollen deposits on another flower of the same individual plant, it is said to be geitonogamy. It usually occurs in plants which show monoecious condition. It is functionally cross-pollination but is generally similar to autogamy because the pollen comes from same plant.

ii. Xenogamy: When the pollen (genetically different) deposits on another flower of a different plant of the same species , it is called as xenogamy.

Contrivances of cross-pollination

The flowers have several mechanisms that promote cross-pollination which are also called **contrivances of cross-pollination or outbreeding** devices. It includes the following.

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